

From Recreational to Elite: Performance Stratification in Marathon Results

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Abstract

Marathon results are often discussed through simple finish-time goals, such as running under three or four hours. These goals are useful, but they do not fully explain how marathon performance is distributed across the whole runner population. This project uses the 2023 Marathon Results dataset to study performance stratification in U.S. marathons. The dataset contains individual results from 641 races and about 429,000 finishers, with runner-level features such as age, gender, finish time, and race name, as well as race-level features such as race date and number of finishers (Rock 2023). We use exploratory data analysis and static visualization to compare performance bands across demographic and race-level contexts. The project focuses on clear visual explanation rather than prediction. The expected contribution is a set of visual summaries that show how recreational, competitive, and near-elite marathon performance differs by age, gender, race size, race timing, and race competitiveness.

I. Introduction

A marathon result is more than a single finishing time. The same finish time can have different meanings depending on the runner's age group, gender, and race context. For example, a sub-4-hour finish may be a strong recreational performance in one group, but it may represent a different competitive level in another group. Similarly, a race with thousands of finishers may have a different performance profile from a small local race.

This project studies marathon performance stratification. We define stratification as the way finishers are distributed across performance groups, from recreational finishers to competitive and near-elite runners. Instead of only reporting average finish time or focusing on race winners, we use visualization to show the structure of the full field.

The project is based on the 2023 Marathon Results dataset, which includes U.S. marathon results from 2023. The main research questions are:

1. How are marathon finishers distributed across performance bands?
2. How do age and gender relate to finish-time distributions?
3. How do race-level features, especially race size and race date, shape performance patterns?

4. Which races appear more competitive based on their share of faster finishers?

The project is designed as an exploratory data visualization study. It does not aim to build a predictive model. Its goal is to make patterns in the dataset easier to see, compare, and explain.

II. Related Work and Visualization Background

Previous research has shown that age and sex are important factors in endurance running performance. Leyk et al. studied age-related changes in marathon and half-marathon performance and showed that demographic variables are important when interpreting endurance results (Leyk et al. 2007). This supports our decision to compare finish-time distributions across age and gender groups.

The project also follows basic principles of quantitative display design. Few argues that graphs and tables should make numerical comparisons clear and efficient (Few 2012). This is important for our project because the dataset is large and contains several kinds of comparisons: runner-level comparisons, demographic comparisons, and race-level comparisons. We therefore use familiar visual forms such as histograms, boxplots, heatmaps, stacked bar charts, and scatterplots. These chart types are chosen because they can show distribution, group difference, and race-level variation without requiring a complex model.

III. Dataset Description

A. Data Source: The primary data source is the 2023 Marathon Results dataset from Kaggle (Rock 2023). It contains two related CSV files. The first file, **Results.csv**, contains runner-level results. The second file, **Races.csv**, contains race-level metadata.

The project is cross-sectional. It studies marathons from 2023 only. This means the dataset is useful for comparing runners and races within one year, but it is not suitable for long-term trend analysis.

B. Runner-Level Features: The runner-level table includes the following fields:

- **Name:** runner name. This is treated as an identifier only. The project focuses on aggregated patterns, not individual runners.
- **Race:** the marathon event name. This field links each runner to a race.
- **Year:** the race year. In this dataset, it is 2023.

- **Gender**: runner gender category. The dataset includes M, F, X, and U.
- **Age**: runner age as a numeric value.
- **Finish**: finish time in seconds.
- **Age Bracket**: age group label, such as 35-39 or 45-49.

The most important runner-level outcome is **Finish**. We convert it from seconds to hours because hours are easier to interpret in a marathon context. We also use **Age**, **Gender**, and **Age Bracket** to compare demographic patterns.

C. Race-Level Features: The race-level table includes:

- **Race**: race name.
- **Year**: race year.
- **Date**: race date.
- **Finishers**: number of finishers in the race.

These fields are important because they let us go beyond age, gender, and performance band. **Race** allows event-level comparison. **Date** allows us to study race month and season. **Finishers** allows us to classify races by size. These race-level features help answer whether different race contexts have different performance profiles.

D. Derived Features: Several derived variables are created for analysis:

- Finish time in hours: **Finish** / 3600.
- Performance band: categories based on finish time.
- Race size group: categories based on the number of finishers.
- Race month and season: derived from **Date**.
- Race-level median finish time.
- Race-level share of sub-3, sub-4, and sub-5 finishers.

These derived variables make the dataset easier to visualize. They also connect individual performance to race-level context.

E. Data Quality and Scope: The dataset needs basic cleaning before analysis. Some records have unknown age brackets. The **Gender** field includes small groups such as X and U, which should be described carefully because the counts are much smaller than the M and F groups. The **Age** field includes unusual values, including negative values, so implausible ages need to be checked. Some races have very small finisher counts, so race-level competitiveness charts should use a minimum race-size threshold.

Finish-time outliers also need review. Extreme times may be valid for walkers or late finishers, but they can strongly affect some charts. For this reason, we do not remove outliers automatically. We first inspect them and then decide whether a chart should show all results or use a focused range.

IV. Methodology

A. Data Cleaning: The analysis will follow a reproducible workflow:

1. Load **Results.csv** and **Races.csv**.
2. Convert finish time from seconds to hours.
3. Parse race date into month and season.
4. Check missing, unknown, and implausible values.
5. Join runner-level records with race-level metadata using **Race** and **Year**.
6. Create race size groups and performance bands.
7. Produce summary tables and static visualizations.

The joined dataset will allow each finisher to be analyzed with both personal features and race context.

B. Performance Bands: Performance bands are used to make finish-time comparisons easier. The initial bands are:

- Elite / near-elite: under 2:45.
- Sub-3: 2:45 to under 3:00.
- Sub-4: 3:00 to under 4:00.
- Sub-5: 4:00 to under 5:00.
- 5+ hours: 5:00 or slower.

These bands are simple and easy to interpret. They are not perfect measures of ability because they do not adjust for age, gender, or course difficulty. However, they are useful for exploratory visualization because many marathon runners and race observers already understand these thresholds.

If time permits, the Boston Cutoff Time Tracker dataset may be used later as a supplementary source (Rock 2025). That dataset would allow us to compare fixed finish-time bands with Boston qualifying standards. It is not part of the current local data package because of its file size.

C. Visualization Design: The project uses static visualizations. Each chart is selected for a specific purpose:

- Histogram or density plot: show the overall shape of finish-time distribution.
- Bar chart: show the number or share of finishers in each performance band.
- Boxplot or violin plot: compare finish-time distributions across age and gender groups.
- Heatmap: show the relationship between age bracket and performance band.
- Stacked bar chart: compare performance-band composition across gender or race size group.
- Scatterplot: compare race size with race competitiveness.
- Ranked bar chart: identify races with high shares of sub-3 or sub-4 finishers.
- Month chart: show how races and finishers are distributed across the 2023 calendar.

The goal is to use charts that are easy to read and easy to compare. We avoid overly complex graphics when a simpler chart can answer the question more clearly.

V. Preliminary Exploration and Planned Results

Initial inspection confirms that the dataset is large enough for both runner-level and race-level visualization. The

runner-level file contains about 429,000 finisher records. The race-level file contains 641 race records. The largest races include major events such as the NYC Marathon, Chicago Marathon, Boston Marathon, LA Marathon, and Honolulu Marathon. At the other end, some races have very small fields.

This wide range of race sizes is important. It means that the project should not only compare individual runners. It should also compare race contexts. A major race with tens of thousands of finishers may have a different performance distribution from a race with fewer than 100 finishers.

A. Overall Performance Distribution: The first result section will show the full distribution of marathon finish times. We expect most finishers to cluster between about 3.5 and 5.5 hours, with a smaller group finishing under 3 hours and a long tail of slower finishers.

Planned figures:

- Figure 1: Histogram of finish time in hours.
- Figure 2: Bar chart of finishers by performance band.

These figures will answer the basic question: how large are the recreational, competitive, and near-elite groups?

B. Demographic Stratification: The second result section will compare age and gender groups. Instead of only reporting average times, we will use distributional charts. This is important because two groups can have similar averages but different spread, skew, or proportions in faster bands.

Planned figures:

- Figure 3: Finish-time distribution by gender.
- Figure 4: Age bracket by performance band heatmap.

These figures will show how demographic features relate to performance structure.

C. Race Size and Race Context: The third result section will focus on race-level features. This section directly addresses the need to discuss features beyond level, age, and gender. We will classify races by number of finishers and compare the performance-band composition of different race size groups.

Planned figures:

- Figure 5: Distribution of race sizes.
- Figure 6: Performance-band composition by race size group.
- Figure 7: Race size versus sub-4 percentage.

These charts will show whether large races and small races have different competitive profiles.

D. Race Date, Month, and Season: The fourth result section will use the `Date` field. Race date is useful because marathon events are not evenly distributed across the year. Some months may contain more races or more finishers.

Month and season may also be related to race type, race size, and participant mix.

Planned figures:

- Figure 8: Number of races by month.
- Figure 9: Number of finishers by month.
- Optional Figure 10: Median finish time by month or season.

This section will show how the race calendar shapes the dataset.

E. Race-Level Competitiveness: The final result section will compare races directly. For each race, we can compute metrics such as median finish time, sub-3 percentage, sub-4 percentage, and sub-5 percentage. To avoid misleading results from very small races, we will filter race-level charts using a minimum finisher threshold.

Planned figures:

- Figure 11: Top races by sub-3 percentage.
- Figure 12: Top races by sub-4 percentage.
- Figure 13: Race finishers versus median finish time.

These charts will identify races that appear more competitive and races that appear more recreational.

VI. Discussion

The expected analysis should show that marathon performance is not evenly distributed. Most finishers are likely to fall into recreational and mid-pack bands, while sub-3 and near-elite finishers make up a much smaller share. Age and gender will likely show different distribution patterns, but these variables are only part of the story.

Race-level context is also important. Race size may affect the performance profile because large races often attract a wider mix of participants, including both competitive runners and recreational runners. Smaller races may show more unstable percentages because a few fast or slow runners can strongly affect the race-level summary. Race date may also matter because the U.S. marathon calendar is concentrated in certain seasons.

The main value of the project is not that it produces one final ranking of runners or races. Its value is that it shows how different views of the same dataset can lead to different interpretations. A fixed threshold such as sub-4 is easy to understand, but it becomes more meaningful when it is viewed by age group, gender, race size, and race context.

VII. Limitations

This project has several limitations. First, the primary dataset covers only U.S. marathons in 2023. It cannot show long-term changes in marathon performance. Second, the dataset does not include weather, course elevation, training history, or split times. These factors may affect performance but cannot be analyzed here. Third, some

fields contain unusual or unknown values, including unknown age brackets and small gender categories. These values need careful treatment in visual summaries.

Race-level analysis also has limitations. Some races have very few finishers, which can make percentages unstable. For example, a small race may appear highly competitive if only a few runners finished and one of them was fast. To reduce this problem, race competitiveness charts should use a minimum finisher threshold.

Finally, fixed performance bands are simple but imperfect. A 3:30 marathon may mean different things for different age and gender groups. Future work could use Boston qualifying standards or age-graded performance metrics to create more adjusted comparisons.

VIII. Conclusion

This project uses data visualization to study marathon performance stratification in 2023 U.S. marathon results. The analysis focuses on how finishers are distributed across recreational, competitive, and near-elite performance bands. It also expands the analysis beyond age and gender by using race-level features such as race name, race date, and number of finishers.

The first version of the project plan shows that the dataset supports several useful visual comparisons. The final paper will use static charts to explain overall finish-time distribution, demographic patterns, race size effects, race calendar patterns, and race-level competitiveness. By combining runner-level and race-level views, the project aims to make marathon performance structure easier to understand than raw result tables or simple averages alone.

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